## Final Project: Executive Banking Dashboard

### Business Scenario

The Head of Retail Banking has asked you to develop an interactive dashboard that enables management to monitor customer activity, account performance, loan portfolio, branch performance, and transaction trends.

You have access to the bank's database and are expected to deliver a professional BI solution.

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# Part 1 – Database Analysis (10%)

Before writing any SQL, understand the database.

### Deliverables

- Identify the purpose of each table.
- Draw an ER Diagram showing the relationships.
- Explain how the tables are connected.
- Identify the fact tables and dimension tables.

The database includes entities such as Branches, Customers, Accounts, Transactions, Loans, Employees, Cards, Transfers, Fees, Loan Payments, Standing Orders, ATMs, and others, with defined foreign-key relationships.

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# Part 2 – SQL (30%)

Write SQL queries to answer business questions.

Examples:

### Customer Analysis

- Total customers
- Customers by city
- Customers by branch

- Monthly customer registrations

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## Account Analysis

- Active vs Closed accounts
- Total balances by branch
- Average account balance
- Savings vs Checking accounts

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## Transaction Analysis

- Monthly transaction volume
- Deposits vs Withdrawals
- Largest transactions
- Most active accounts

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## Loan Analysis

- Active loans
- Paid loans
- Loan amounts by type
- Average interest rate
- Outstanding loans

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## Card Analysis

- Active cards
- Credit vs Debit cards
- Expired cards

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## Branch Performance

- Customers per branch
- Employees per branch
- Accounts per branch

- Total balances

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# Part 3 – Excel (10%)

Export the SQL results into Excel.

Perform:

- Data validation
- Duplicate checking
- Missing value checks
- Formatting
- Pivot Tables
- Basic charts

Prepare the dataset for Power BI.

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# Part 4 – Power BI (35%)

Create a professional dashboard.

## Page 1 – Executive Overview

KPIs

- Total Customers
- Total Accounts
- Total Balance
- Total Transactions
- Total Loans
- Total Branches

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## Page 2 – Customer Insights

Visuals

- Customers by Branch
- Customers by City
- Monthly New Customers
- Customer Distribution

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## Page 3 – Accounts

Visuals

- Account Types
- Active vs Closed
- Balance by Branch
- Top 10 Customers by Balance

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## Page 4 – Transactions

Visuals

- Deposits vs Withdrawals
- Monthly Trend
- Largest Transactions
- Average Transaction Amount

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## Page 5 – Loans

Visuals

- Loans by Type
- Loan Status
- Loan Amount by Branch
- Average Interest Rate

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## Page 6 – Branch Performance

Visuals

- Branch Ranking
- Employees
- Customers
- Accounts
- Total Deposits

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### Required Slicers

- Date
- Branch
- City
- Customer
- Loan Type
- Account Type

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# Part 5 – DAX (10%)

Create at least 15 measures.

Examples

- Total Customers
- Total Accounts
- Total Balance
- Active Accounts
- Average Balance
- Total Loan Amount
- Average Interest Rate
- Transaction Count
- Deposit Amount
- Withdrawal Amount
- Running Total Balance
- YTD Transactions
- Monthly Growth %
- Rank Branch
- Average Loan Size

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# Part 6 – Business Insights (5%)

Write a one-page report answering questions such as:

- Which branch performs the best?
- Which account type is most popular?
- Which loan type generates the highest portfolio?
- Are customer registrations increasing?
- Which branches need attention?
- What recommendations would you make to management?

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# Evaluation Rubric

| Section | Weight |
|---|---|
| Database Understanding | 10% |
| SQL Queries | 30% |
| Excel Preparation | 10% |
| Power BI Dashboard | 35% |
| DAX Measures | 10% |
| Business Insights | 5% |

### Bonus Challenge (+10%)

Without additional guidance, create **one extra dashboard page** that provides value to management using any of the remaining tables (e.g., Fees, Cards, ATMs, Standing Orders, Audit Log, or Transfers). Explain why you chose it and what business decisions it supports.